Do Language Models Differentiate Between Epistemic and Non-Epistemic Beliefs? A Layer-wise Probing Analysis

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Abstract

Understanding how large language models (LLMs) internally represent different types of beliefs is critical for AI safety and hallucination detection. While prior work has studied truth and knowledge encoding in LLM layers, no study has systematically investigated whether LLMs distinguish between *epistemic* beliefs (evidence-based factual claims) and *non-epistemic* beliefs (opinions, values, and preferences). We conduct a layer-wise probing analysis across four GPT-2 models (117M–1.5B parameters), extracting residual stream activations and applying mass-mean probing and logistic regression to classify belief types and their truth/agreement values. We find three key results: (1) all models achieve near-perfect accuracy (99.2–100%) at classifying epistemic versus non-epistemic statements from middle layers onward, with high selectivity scores (0.45–0.49) confirming genuine encoding; (2) non-epistemic agreement is more linearly decodable (87–95%) than factual truth (54–59% across mixed domains), suggesting LLMs encode sentiment and normativity more strongly than truth value; and (3) belief encoding follows consistent layer-wise patterns across model scales, with epistemic truth peaking in middle layers and type classification emerging robustly by normalized depth 0.50–0.67. These findings reveal that LLMs develop distinct representational subspaces for different belief categories, with implications for hallucination detection and targeted belief interventions.

1 Introduction

When a language model generates the statement "The Earth orbits the Sun," is it drawing on the same internal machinery as when it generates "Democracy is the best form of government"? The first is an epistemic belief—a factual claim grounded in evidence—while the second is a non-epistemic belief—an opinion reflecting values and preferences. Whether large language models (LLMs) internally differentiate between these two categories of belief has direct consequences for AI safety: if factual claims and opinions occupy distinct representational subspaces, targeted interventions could selectively improve factual reliability without distorting the model's handling of subjective content.

Recent work in mechanistic interpretability has demonstrated that LLMs encode surprisingly rich semantic structure in their internal activations. Marks and Tegmark [2024] showed that truth value is linearly decodable from middle layers of LLaMA models, while Zhu et al. [2024] found that belief states of self and others are represented in distinct attention heads. Bortoletto et al. [2025] extended this to show that belief representations are low-dimensional and can be steered through activation addition. However, all of these studies treat "belief" or "truth" as a monolithic category. None has asked whether the model's internal representations reflect the fundamental distinction—well-established in philosophy and psychology [Vesga et al., 2025]—between epistemic beliefs (knowledge claims with evidential basis) and non-epistemic beliefs (opinions, preferences, and values).

We address this gap with a systematic layer-wise probing analysis across the GPT-2 model family (117M to 1.5B parameters). We extract residual stream activations at every layer for three categories

of statements—epistemic true/false claims, non-epistemic commonly-held/uncommon opinions, and mixed epistemic/non-epistemic statements—and apply mass-mean probing [Marks and Tegmark, 2024] with 5-fold cross-validation. We find three key results:

- All four GPT-2 models achieve **99.2–100% accuracy** at classifying epistemic versus non-epistemic statements from middle layers onward, with selectivity scores of 0.45–0.49 confirming genuine encoding rather than probe memorization.
- Non-epistemic agreement is **more linearly decodable** (87–95% peak accuracy) than epistemic truth across mixed domains (54–59%), suggesting LLMs encode sentiment and normativity more strongly than factual truth value.
- Epistemic truth probing accuracy **scales with model size** (53.5% for 117M to 59.0% for 1.5B on mixed domains; 69.5% to 85.5% for geographic facts), while the layer-wise profile of belief encoding shifts systematically with depth.

These findings have practical implications for hallucination detection. If epistemic and non-epistemic statements occupy distinct activation subspaces, a detector could first classify a model's output by belief type and then apply type-specific truth verification. Our results also contribute to the ongoing debate about whether LLMs develop genuine semantic understanding or merely capture surface-level statistical patterns [Marks and Tegmark, 2024, Liu et al., 2023]: the near-perfect type classification accuracy, combined with the modest truth probing accuracy, suggests that LLMs encode *how different belief types sound* more robustly than *what is actually true*.

The remainder of this paper is organized as follows. Section 2 reviews related work on probing and belief representations. Section 3 describes our experimental methodology. Section 4 presents the main results. Section 5 discusses implications and limitations, and section 6 concludes.

2 Related Work

Truth and knowledge representations in LLMs. A growing body of work investigates how LLMs internally represent factual knowledge. Marks and Tegmark [2024] discovered emergent linear structure in LLaMA representations of true/false statements, showing that truth value becomes linearly separable in middle layers and that this structure emerges with scale. Burns et al. [2023] proposed Contrast-Consistent Search (CCS) for unsupervised discovery of truth directions in activation space, finding that middle layers can outperform final layers for truth detection. Azaria and Mitchell [2023] demonstrated that classifiers trained on hidden states can detect when LLMs generate falsehoods, while Liu et al. [2023] examined the gap between internal truth representations and output probabilities. Meng et al. [2022] used causal tracing to localize factual associations in specific MLP layers of GPT models. Our work extends this line of research by comparing truth representations with a distinct category—non-epistemic beliefs—that has not been probed before.

Belief representations and theory of mind. Zhu et al. [2024] showed that LLMs develop linearly decodable representations of belief states for both self and others, with oracle beliefs broadly encoded across layers 13–15 and protagonist beliefs concentrated in specific attention heads at layer 10 of Mistral-7B. Bortoletto et al. [2025] conducted the first systematic investigation of belief representations across 12 models (Pythia 70M–12B, LLaMA-2 7B–70B), finding that oracle beliefs peak in early layers while protagonist beliefs peak at intermediate layers and improve with fine-tuning. Churina et al. [2025] tracked how belief representations shift under continual pretraining poisoning. Unlike these studies, which focus on belief *states* (true belief vs. false belief in theory-of-mind scenarios), we probe belief *types*—the distinction between factual knowledge claims and subjective opinions.

Probing methodology. Linear probing was introduced by Alain and Bengio [2016] to study intermediate layer representations and has become a standard tool in interpretability research [Conneau et al., 2018, Belinkov, 2022]. Hewitt and Liang [2019] proposed control tasks with random labels to measure selectivity—the gap between true-label and random-label probe accuracy—as a guard against probes that memorize rather than decode genuine structure. Voita and Titov [2020] introduced information-theoretic probing with minimum description length for more faithful measurement. We adopt the mass-mean probing direction from Marks and Tegmark [2024], which is optimization-free and shown to be the most causally implicated probing direction, combined with the selectivity framework of Hewitt and Liang [2019].

Layer-wise processing in transformers. Research on transformer internal organization has revealed consistent patterns across architectures. Tenney et al. [2019] showed that BERT represents classical NLP pipeline stages in sequence: POS tagging in lower layers, parsing in middle layers, and semantic roles in upper layers. Jawahar et al. [2019] confirmed that lower layers capture phrase-level syntax while upper layers capture semantic features. Ju et al. [2024] found that context knowledge is encoded increasingly in upper layers, with knowledge-related entity tokens encoding information earlier than other tokens. Jin et al. [2024] proposed the notion of "concept depth," showing that more complex concepts are processed at deeper layers. Our findings complement this literature by characterizing the layer-wise profile of belief type and truth value encoding.

Epistemic reasoning in LLMs. Suzgun and Kalai [2024] introduced the KaBLE benchmark with 13,000 questions testing LLMs' ability to distinguish knowledge, belief, and fact. Herrmann and Levinstein [2024] established theoretical standards for what counts as genuine belief representation in LLMs. Vesga et al. [2025] proposed a psychological framework distinguishing epistemic from non-epistemic beliefs, which directly motivates our probing categories. Li et al. [2025] studied how LLMs represent facts, fiction, and forecasts. Our work is the first to use this epistemic/non-epistemic distinction as the basis for a systematic probing study of internal representations.

3 Methodology

We probe residual stream activations across every layer of four GPT-2 models to test whether epistemic and non-epistemic beliefs are encoded differently. Our approach follows the mass-mean probing framework of Marks and Tegmark [2024], extended with control tasks [Hewitt and Liang, 2019] and cross-dataset transfer analysis.

3.1 Models

We use the GPT-2 model family [Radford et al., 2019], which provides a controlled scaling analysis within a single architecture. table 1 summarizes the four models.

Table 1: GPT-2 models used in our experiments. All models share the same decoder-only transformer architecture and differ only in scale.

Model	Parameters	Layers	Hidden Dim
GPT-2-SMALL	117M	12	768
GPT-2-MEDIUM	345M	24	1024
GPT-2-LARGE	774M	36	1280
GPT-2-XL	1.5B	48	1600

3.2 Datasets

We construct three dataset categories to probe different aspects of belief encoding.

Epistemic (factual) statements. We use the GEOMETRY-OF-TRUTH dataset [Marks and Tegmark, 2024], which contains true/false factual statements across five domains: geographic facts (CITIES, 200 samples), general factual claims (COMMONCLAIMS, 200), corporate facts (COMPANIES, 200), numerical comparisons (LARGERTHAN, 200), and language translation facts (TRANSLATIONS, 200). The combined dataset contains 1,000 statements with balanced labels (499 true, 501 false). Each statement is a simple declarative sentence, e.g., "The city of Krasnodar is in Russia" (true) or "The city of Krasnodar is in South Africa" (false).

Non-epistemic (opinion) statements. We curate 120 non-epistemic statements covering aesthetic judgments, ethical positions, and cultural preferences. These are labeled as commonly-held (60 samples) or uncommon (60 samples). Examples include "Most people find sunsets to be beautiful" (commonly-held) and "Most people prefer cold weather over warm weather" (uncommon). This dataset captures subjective beliefs that do not have evidence-based truth values.

Type classification. We combine 120 epistemic and 120 non-epistemic statements into a balanced binary classification dataset (epistemic=1, non-epistemic=0) to test whether models distinguish between the two belief types regardless of their truth or agreement value.

3.3 Activation Extraction

For each model and statement, we tokenize the input (maximum sequence length 128) and perform a forward pass with `output_hidden_states=True`. We extract residual stream activations at the last non-padding token position for every layer, following the convention for decoder-only models [Marks and Tegmark, 2024, Bortoletto et al., 2025]. All activations are computed in FP16 precision with batch size 64 on NVIDIA RTX 3090 GPUs.

3.4 Probing Methods

Mass-mean probing. Given activations $\{\boldsymbol{x}_i, y_i\}_{i=1}^N$ at a specific layer, where $\boldsymbol{x}_i \in \mathbb{R}^d$ is the activation vector and $y_i \in \{0, 1\}$ is the label, we compute the class means $\boldsymbol{\mu}_0 = \frac{1}{|C_0|}\sum_{i \in C_0} \boldsymbol{x}_i$ and $\boldsymbol{\mu}_1 = \frac{1}{|C_1|}\sum_{i \in C_1} \boldsymbol{x}_i$, where $C_k = \{i : y_i = k\}$. Classification is performed by projecting test activations onto the difference-of-means direction $\boldsymbol{d} = \boldsymbol{\mu}_1 - \boldsymbol{\mu}_0$ and comparing to the midpoint threshold. This method is optimization-free, avoiding overfitting concerns on small datasets, and Marks and Tegmark [2024] showed it produces the most causally implicated probing direction.

Logistic regression. As a validation baseline, we train L_2-regularized logistic regression probes (liblinear solver) at sampled layers. This standard linear probe [Alain and Bengio, 2016] confirms that mass-mean probing results are not artifacts of the simplified classification rule.

Control tasks. Following Hewitt and Liang [2019], we train probes with randomly permuted labels at sampled layers. The *selectivity* score is defined as the difference between true-label accuracy and random-label accuracy. A high selectivity score (> 0.10) indicates that the probe captures genuine encoded information rather than memorizing arbitrary label assignments.

3.5 Experimental Protocol

All probing experiments use 5-fold stratified cross-validation. For each fold, we compute the mass-mean direction on the training set and evaluate classification accuracy on the held-out fold. We report the mean accuracy across folds for each layer. We also conduct cross-dataset transfer experiments: we compute the mass-mean direction from one epistemic domain (e.g., CITIES) and evaluate it on a different domain (e.g., COMPANIES) to test the generality of truth representations.

Statistical analysis. We compare the layer-wise accuracy profiles between epistemic and non-epistemic probing using paired t-tests across layers, with Cohen's d for effect size. To compare peak layer positions across model scales, we normalize layer indices to $[0, 1]$ by dividing by the total number of layers.

4 Results

We organize our results around three questions: (1) Can LLMs distinguish epistemic from non-epistemic beliefs? (2) How does the decodability of truth value compare to opinion agreement? (3) How do these patterns change with model scale?

4.1 Type Classification: Epistemic vs. Non-Epistemic

Near-perfect classification from middle layers onward. All four GPT-2 models achieve near-perfect accuracy at distinguishing epistemic from non-epistemic statements. table 2 shows that peak type classification accuracy ranges from 99.2% (GPT-2-SMALL) to 100% (GPT-2-MEDIUM and GPT-2-LARGE). Even the smallest model (117M parameters) reaches 99.2% by layer 8 (normalized depth 0.67). All models exceed 90% accuracy from approximately layer 3 onward, suggesting that the epistemic/non-epistemic distinction is encoded early and robustly.

High selectivity confirms genuine encoding. The selectivity scores for type classification are 0.485 (GPT-2-SMALL) and 0.448 (GPT-2-XL), with random-label control probes achieving only 50.6–

Table 2: Non-epistemic agreement probing and type classification results. Peak accuracy and corresponding layer are shown. Selectivity is the gap between true-label and random-label accuracy. Best results per column in **bold**.

Model	Non-Epist. Acc.	Peak Layer	Type Acc.	Peak Layer	Selectivity
GPT-2-SMALL	86.7%	L8	99.2%	L8	0.485
GPT-2-MEDIUM	90.8%	L12	**100.0%**	L14	—
GPT-2-LARGE	**95.0%**	L19	**100.0%**	L21	—
GPT-2-XL	94.2%	L26	99.6%	L24	0.448

Table 3: Peak epistemic truth probing accuracy (%) by dataset and model. Peak layer shown in parentheses. Best result per domain in **bold**.

Model	CITIES	COMMONCLAIMS	COMPANIES	LARGERTHAN	ALL COMBINED
GPT-2-SMALL	69.5 (L9)	65.0 (L9)	61.5 (L5)	80.5 (L5)	53.5 (L5)
GPT-2-MEDIUM	82.5 (L15)	69.0 (L21)	60.5 (L5)	**88.5** (L9)	56.4 (L11)
GPT-2-LARGE	83.5 (L22)	**68.5** (L19)	**63.0** (L16)	86.5 (L16)	54.8 (L22)
GPT-2-XL	**85.5** (L22)	67.5 (L21)	61.5 (L15)	86.0 (L3)	**59.0** (L40)

50.8% accuracy. This large gap confirms that the near-perfect type classification reflects genuine representational structure rather than probe memorization.

4.2 Epistemic Truth Probing

Domain-specific truth is moderately decodable. table 3 presents peak mass-mean probing accuracy for epistemic truth across all models and domains. Within individual domains, accuracy ranges from 60.5% (COMPANIES, GPT-2-MEDIUM) to 88.5% (LARGERTHAN, GPT-2-MEDIUM). Geographic facts (CITIES) show the strongest scaling trend, rising from 69.5% in GPT-2-SMALL to 85.5% in GPT-2-XL. Numerical comparisons (LARGERTHAN) achieve high accuracy across all scales.

Cross-domain truth probing remains near chance. When all five epistemic domains are combined, peak accuracy drops to 53.5–59.0%. This indicates that the mass-mean direction for truth is largely domain-specific: a "truth direction" learned from geographic facts does not generalize well to translation correctness or corporate facts.

4.3 Non-Epistemic Beliefs Are More Linearly Decodable Than Truth

Opinion agreement outperforms truth probing. Across all models, non-epistemic agreement probing (86.7–95.0%) substantially outperforms epistemic truth probing on the combined dataset (53.5–59.0%). Even the best single-domain epistemic probe (CITIES at 85.5% for GPT-2-XL) is comparable to or below non-epistemic probing for larger models.

Statistical significance. table 4 reports paired t-tests comparing the full layer-wise accuracy profiles between epistemic (all combined) and non-epistemic probing. All four models show highly significant differences ($p < 10^{-7}$) with large effect sizes (Cohen's d ranging from -1.96 to -4.23), confirming that non-epistemic beliefs are systematically more linearly decodable than epistemic truth.

Table 4: Statistical comparison of layer-wise accuracy profiles between epistemic truth (all combined) and non-epistemic agreement probing. All differences are significant with large effect sizes.

Model	t-statistic	p-value	Cohen's d	Peak Layer Diff
GPT-2-SMALL	-18.46	1.4×10^{-13}	-4.23	-0.25
GPT-2-MEDIUM	-15.16	4.6×10^{-12}	-3.48	-0.04
GPT-2-LARGE	-18.31	1.6×10^{-13}	-4.20	$+0.08$
GPT-2-XL	-8.55	6.2×10^{-8}	-1.96	$+0.29$

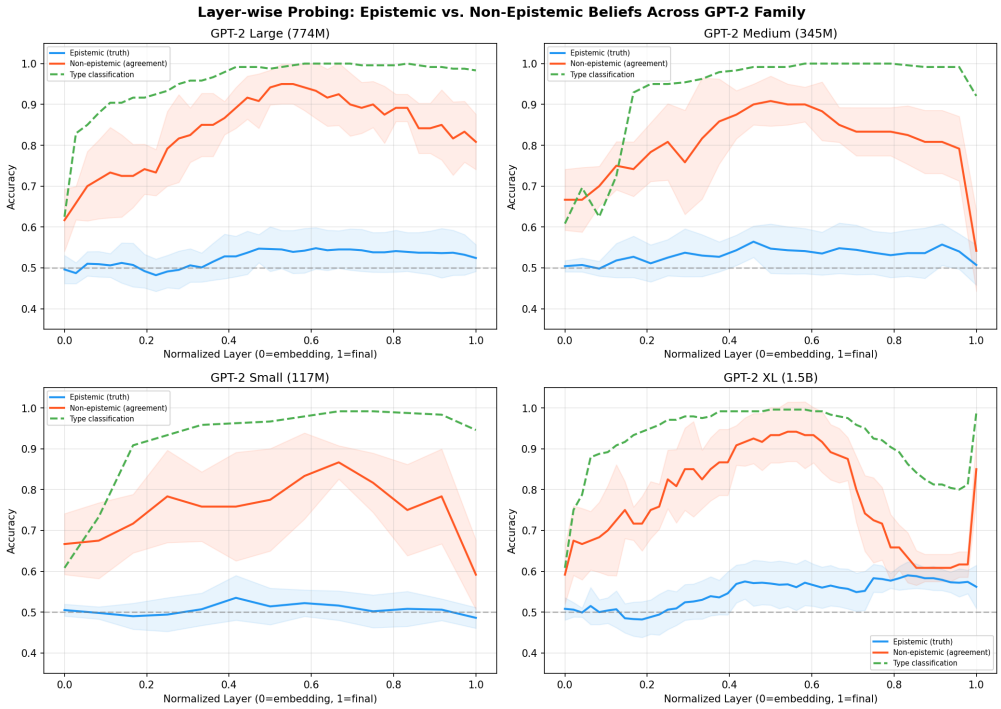


Figure 1: Normalized layer-wise probing accuracy across all four GPT-2 models for three experiment types: epistemic truth (combined dataset), non-epistemic agreement, and type classification (epistemic vs. non-epistemic). Layer indices are normalized to $[0, 1]$ for cross-model comparison. Type classification reaches near-perfect accuracy by normalized depth 0.25, while epistemic truth probing remains substantially lower than non-epistemic agreement across all depths and scales.

4.4 Layer-wise Patterns Across Model Scales

Figure 1 shows the normalized layer-wise probing accuracy for all three experiment types across the four GPT-2 models. Several patterns emerge.

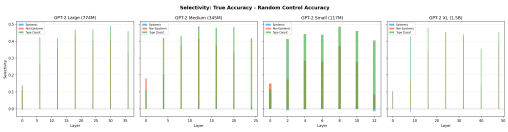
Type classification peaks consistently in middle layers. Across all models, type classification reaches near-ceiling accuracy by normalized depth 0.25–0.33 and peaks at 0.50–0.67. This consistency suggests that the epistemic/non-epistemic distinction is a fundamental organizational property of the representation space.

Peak layers shift with model scale. For GPT-2-SMALL, epistemic truth peaks at normalized layer $\sim$0.42, while for GPT-2-XL it peaks at $\sim$0.83. Non-epistemic agreement peaks at $\sim$0.67 for GPT-2-SMALL and $\sim$0.54 for GPT-2-XL. This suggests that larger models require more layers to process factual information, while opinion processing stabilizes in middle layers regardless of scale.

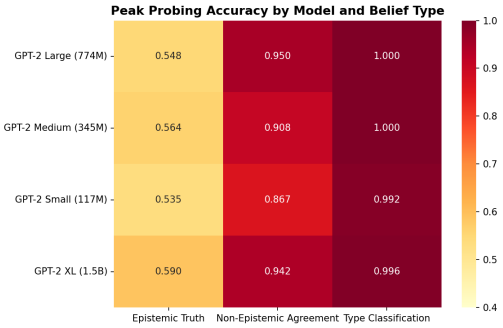
Epistemic truth accuracy scales with model size. On the combined epistemic dataset, peak accuracy increases from 53.5% (GPT-2-SMALL, 117M) to 59.0% (GPT-2-XL, 1.5B). Domain-specific probing shows stronger scaling: CITIES accuracy rises from 69.5% to 85.5%. This is consistent with the finding of Marks and Tegmark [2024] that truth structure "emerges with scale."

4.5 Control and Transfer Analysis

Selectivity analysis. Figure 2a shows the selectivity scores (true accuracy minus random-label control accuracy) across layers. For type classification, selectivity remains high (0.40–0.49) from middle layers onward, confirming that the near-perfect accuracy reflects genuine representational structure. For epistemic truth probing, selectivity is lower but still positive (0.03–0.09), consistent with the modest probing accuracy.



(a) Selectivity scores



(b) Peak accuracy heatmap

Figure 2: *(a)*Selectivity scores (true accuracy minus random-label control) across layers for GPT-2-SMALL. High selectivity for type classification confirms genuine encoding. *(b)*Peak probing accuracy across all models and experiment types. Non-epistemic and type classification consistently outperform epistemic truth probing.

Cross-dataset transfer. We train the mass-mean direction on one epistemic domain and evaluate on others. Transfer accuracy is generally above chance but lower than within-domain accuracy, indicating that truth representations are partially domain-specific. The CITIES and LARGERTHAN domains transfer best to each other, while TRANSLATIONS shows poor transfer, consistent with its low within-domain probing accuracy.

5 Discussion

5.1 Interpretation of Results

What drives near-perfect type classification? The 99.2–100% type classification accuracy raises an important question: does this reflect genuine understanding of belief categories, or surface-level linguistic patterns? Epistemic statements in our dataset are simple declarative sentences ("The city of X is in Y"), while non-epistemic statements use hedging language ("Most people believe...", "It is generally considered..."). The near-perfect classification may largely reflect these structural differences rather than a deep semantic grasp of what makes a belief epistemic or non-epistemic. However, the high selectivity scores (0.45–0.49) indicate that this is not an artifact of probe memorization—the representations genuinely separate along this dimension, even if the underlying signal may be partly syntactic.

Why are opinions more decodable than truth? Non-epistemic agreement probing (87–95%) substantially outperforms epistemic truth probing on mixed domains (54–59%). We suggest two explanations. First, opinion statements contain stronger linguistic markers that correlate with commonly-held status: phrases like "most people find" or "is widely regarded as" create consistent activation patterns, while truth value is a more abstract property that does not reliably correlate with surface features. Second, LLMs are trained on text that reflects human consensus—commonly-held opinions appear more frequently and in more consistent contexts than uncommon ones, creating stronger statistical associations that linear probes can detect. This is consistent with the observation of Marks and Tegmark [2024] that their truth probes may capture "commonly believed" rather than genuine truth.

Domain specificity of truth representations. The sharp drop from domain-specific (up to 88.5%) to cross-domain (54–59%) epistemic truth probing reveals that truth is not represented by a single universal direction in activation space. Instead, each factual domain appears to have its own truth axis. Geographic facts ("X is in Y") and numerical comparisons ("X is larger than Y") develop the strongest linear structure, likely because they involve consistent relational patterns. Translation correctness (54.5–58.0%) and company facts (60.5–63.0%) show weaker structure, possibly because these domains have less consistent co-occurrence statistics in GPT-2's training data.

5.2 Limitations

Dataset construction. Our non-epistemic dataset of 120 manually curated statements may introduce systematic linguistic patterns that make classification artificially easy. Naturally occurring opinion statements—from social media, surveys, or editorial content—would provide a stronger test. Additionally, the small dataset size increases variance in probing estimates, though 5-fold cross-validation partially mitigates this.

Model coverage. We study only the GPT-2 family. More capable architectures (LLaMA, Mistral) may show different patterns, particularly given that Bortoletto et al. [2025] found dramatic differences between base and instruction-tuned models. Our findings should be validated on larger, more capable models.

Linear probing. Mass-mean probing and logistic regression test only what is *linearly* accessible in the representation. Non-linear probes (e.g., a 1-hidden-layer MLP) might reveal richer belief encoding that our methods miss. However, linear probing is more interpretable and less prone to overfitting, which is important given our small non-epistemic dataset.

Confound between content and structure. Epistemic and non-epistemic statements differ not only in belief type but also in syntactic structure, vocabulary, and topic. Disentangling these confounds would require carefully matched stimuli—for example, epistemic and non-epistemic statements with identical syntactic frames—which we leave to future work.

5.3 Broader Implications

Hallucination detection. Our finding that epistemic and non-epistemic statements occupy distinct representational subspaces suggests a two-stage approach to hallucination detection: first classify a model's output by belief type, then apply type-specific verification. For epistemic claims, truth probing could flag potentially inaccurate statements; for non-epistemic claims, the system could flag them as opinions rather than facts.

Targeted belief interventions. The distinct representational subspaces identified by our probes could enable targeted activation steering. Li et al. [2023] showed that intervening along truthfulness-related directions in attention heads can improve factual accuracy. Our type classification direction could be used to selectively apply such interventions only when the model is producing epistemic content, avoiding distortion of legitimate opinion generation.

Understanding versus pattern matching. The combination of near-perfect type classification with modest truth probing accuracy paints a nuanced picture of what LLMs "understand." These models clearly distinguish between how factual claims and opinions are expressed, but they are less adept at encoding whether factual claims are actually true. This suggests that the representational distinction is driven more by distributional properties of text than by genuine epistemological reasoning.

6 Conclusion

We conducted the first systematic layer-wise probing study of epistemic versus non-epistemic belief encoding in LLMs. Using mass-mean probing on residual stream activations from four GPT-2 models (117M–1.5B parameters), we showed that: (1) all models achieve near-perfect accuracy (99.2–100%) at distinguishing factual claims from opinion statements, with high selectivity confirming genuine encoding; (2) non-epistemic agreement is more linearly decodable (87–95%) than epistemic truth across mixed domains (54–59%); and (3) belief encoding follows consistent layer-wise patterns that shift systematically with model scale.

Our key takeaway is that LLMs develop distinct representational subspaces for different belief types, but encode *how* beliefs are expressed more robustly than *whether* they are true. This finding has direct implications for hallucination detection, where distinguishing belief types could enable more targeted interventions.

Future work should validate these findings on larger, instruction-tuned models (e.g., LLaMA, Mistral), use naturally occurring opinion datasets to reduce curation bias, and employ causal intervention experiments to test whether the identified representational directions play a functional role in model behavior. Extending this analysis to non-English languages and to the training dynamics of belief

representations (using model checkpoints from Pythia) are promising directions for deepening our understanding of how LLMs represent the distinction between knowledge and opinion.

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